

Problem

Determine the inverse Fourier transforms of:

(a) $F(\omega) = 4\delta(\omega + 3) + \delta(\omega) + 4\delta(\omega - 3)$

(b) $G(\omega) = 4u(\omega + 2) - 4u(\omega - 2)$

(c) $H(\omega) = 6 \cos 2\omega$

Step-by-step solution

Step 1 of 3

$$\begin{aligned} \text{(A)} \quad f(t) &= F^{-1}[\delta(\omega)] + F^{-1}[us(\omega + 3) + us(\omega - 3)] \\ &= \frac{1}{2\pi} + \frac{4 \cos 3t}{\pi} = \frac{1}{2\pi}(1 + 8 \cos 3t) \end{aligned}$$

Step 2 of 3

$$\begin{aligned} \text{(B)} \quad \text{if} \quad h(t) &= u(t + 2) - u(t - 2) \\ H(\omega) &= \frac{2 \sin 2\omega}{\omega} \\ G(\omega) &= 4H(\omega) \rightarrow g(t) = \frac{1}{2\pi} \cdot \frac{8 \sin 2t}{t} \\ g(t) &= \frac{4 \sin 2t}{\pi t} \end{aligned}$$

Step 3 of 3

$$\begin{aligned} \text{(C)} \quad \text{since} \quad \cos(at) &= \pi\delta(\omega + a) + \pi\delta(\omega - a) \\ \text{using the Reversal property,} \\ 2\pi \cos 2\omega &\leftrightarrow \pi\delta(t + 2) + \pi\delta(t - 2) \\ \text{or,} \quad F^{-1}[6 \cos 2\omega] &= 3\delta(t + 2) + 3\delta(t - 2) \end{aligned}$$